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Design Challenges and Techniques for 5nm FinFET CMOS Analog/Mixed-Signal Circuits

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***Abstract*— Recent CMOS scaling incessantly focuses on improving density, speed, area, and power consumption in digital circuits which is increasingly less applicable to analog/mixed-signal circuits. This paper proposes various practical techniques to overcome technology challenges in several common analog/mixed-signal circuits in 5nm CMOS. Issues addressed include high leakage in dummy devices due to continuous active area layout style, large parasitics in ring oscillators, and gate-induced drain leakage in SAR ADC sampling switches.**

Keywords—5nm finFET technology, leakage currents, gate-induced drain leakage, ring oscillators

I. Introduction

CMOS technology scaling has provided many advantages to the latest IC designs. Despite asymptotic physical feature scaling, design/technology co-optimization in the most advanced nodes continues to offer improved device area density, speed, area, and power to enable unprecedented levels of system performance and integration. Scaling benefits digital circuits but plague analog/mixed-signal circuits with many challenges. For instance, MOS threshold voltage (V_T) is not scaling as aggressively as supply voltage. The scaling of the transistor gate length contributes to leakage. Furthermore, layout rules have become ever more restrictive to ensure manufacturing yield and contain device variation. A plethora of layout techniques have addressed some of these challenges [1]–[4]. For example, [1] proposes a method to reduce leakage currents using intra-gate biasing. The width and length of the transistor gate have been adjusted to increase VT and reduce leakage, but this scheme is not applicable as jogged active areas are discouraged in 5nm. Inverter layout techniques in [2] reduce inverter propagation delays by decreasing the routing length of the input and output nodes. Ref [4] presents a multi-finger layout technique for better matching and performance. This technique reduces the transistor's intrinsic resistance but does little to reduce parasitic resistance and capacitance.

This paper addresses several common issues faced when porting analog/mixed-signal designs from older nodes to 5nm CMOS: leakage in dummy transistors, interconnect parasitics in ring oscillators, and gate-induced drain leakage in SAR ADC sampling switches. We introduce layout techniques to reduce leakage as well as parasitic resistance and capacitance, especially significant in the middle-end-of-line and lower back-end-of-line. Section II describes the impact of leakage current flowing in mandatory dummy transistors around the active devices. Section III addresses the impact of parasitics in high-performance ring oscillators. Section IV describes the impact of high gate-induced drain leakage in SAR ADC sampling switches.

II. Overcoming Leakage to Dummy Transistors

Continuous active area layout styles are increasingly common for device density and performance improvement. This layout style establishes a stress plateau where the mobility of devices is stable and maximal [5]. Electrically deactivated dummy devices are inserted in between two active devices to maintain continuity in the active area. The NMOS and PMOS dummy device terminals are connected to ground and supply, respectively. These dummy devices around the active devices contribute to leakage, especially at the fast process corner and at high temperatures, thereby degrading the design performance.

Two possibilities exist for connecting dummy transistors around the active devices. The first is to connect the dummy devices (MD) in the centre as shown in Fig. 1. The gates of the MOS and PMOS dummies are connected to VSS and VDD, respectively, making them electrically deactivated. When Sig1 and Sig2 voltages are different, subthreshold leakage can flow. Another possibility is connecting the dummy transistors at the end. The Gate/Source/Drain of NMOS and PMOS dummies are connected to VSS and VDD, respectively. Leakage flows from Sig1/Sig2 to VDD/VSS through the dummies as shown in Fig. 2.

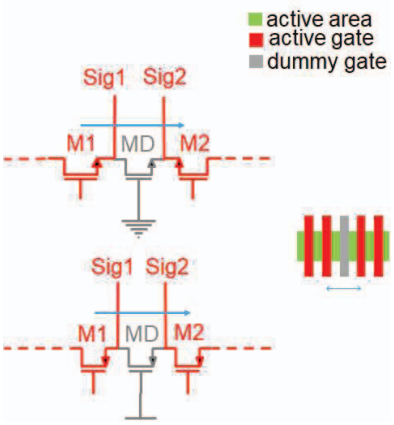


Fig.1. Dummy transistors in the center of a continuous active area.

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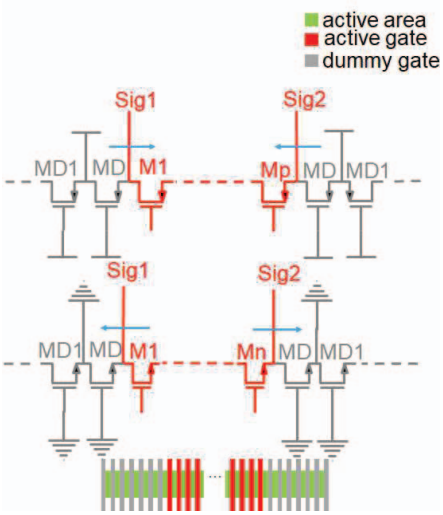


Fig.2. Dummy transistors at the end of an active area.

Fig. 3 illustrates two methods to reduce the leakage currents to the dummy transistors. In the first method, the diffusion of dummy transistors is connected to signal nets Sig1/Sig2 as shown by the green dashed line. This method adds more capacitance to the signal nets, which reduces the speed of operation. This results in a lower frequency of changing the trim bits in the resistor ladder. The second method is to insert a larger number of dummy devices. By floating the intermediate diffusion nodes of the series dummies, the effective channel length of the dummy device tied to Sig1/Sig2 is significantly increased, thereby dramatically reducing leakage. The length of dummy devices will be N times the length of a single transistor connected as a dummy device, where N is the total number of transistors.

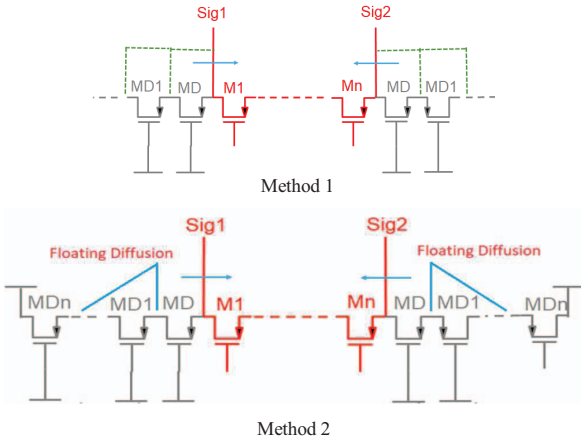


Fig. 3. Methods to reduce leakage drawn by adjacent dummy devices.

To illustrate the significance of leakage to dummies and our proposed methods to reduce this leakage, consider a resistor ladder as shown in Fig 4 for generating a reference voltage (V_{ref}) that dictates the frequency of a free-running oscillator (FRO). The current flowing through the resistor ladder is 2 µA. Six bits exist for fine trimming, so 64 switches are tapped along the ladder to select the desired V_{ref}.

$$V_{ref} = \frac{VDD_CORE*150K}{(COARSEBIT*10.7K+FINEBIT*538+75K)}$$

Dummy devices surround the active devices to protect the active devices from variations due to density gradients [6]. The leakage current ($I_{leakage}$) at fast processes and hot corners drawn by these dummies becomes a significant portion of the total current consumed by the circuit. As shown in Table I, the leakage current in simulations becomes 14% of the active current flowing in the resistor ladder, subsequently disturbing V_{ref} at the hot temperature as shown in Table II. This results in an unacceptably large 10% frequency deviation that violates the design specification. With the improved layout as per the Method II shown in Fig 3, the leakage current reduces to only 2.9% of the active ladder current, causing less than 1-mV of V_{ref} deviation across temperature.

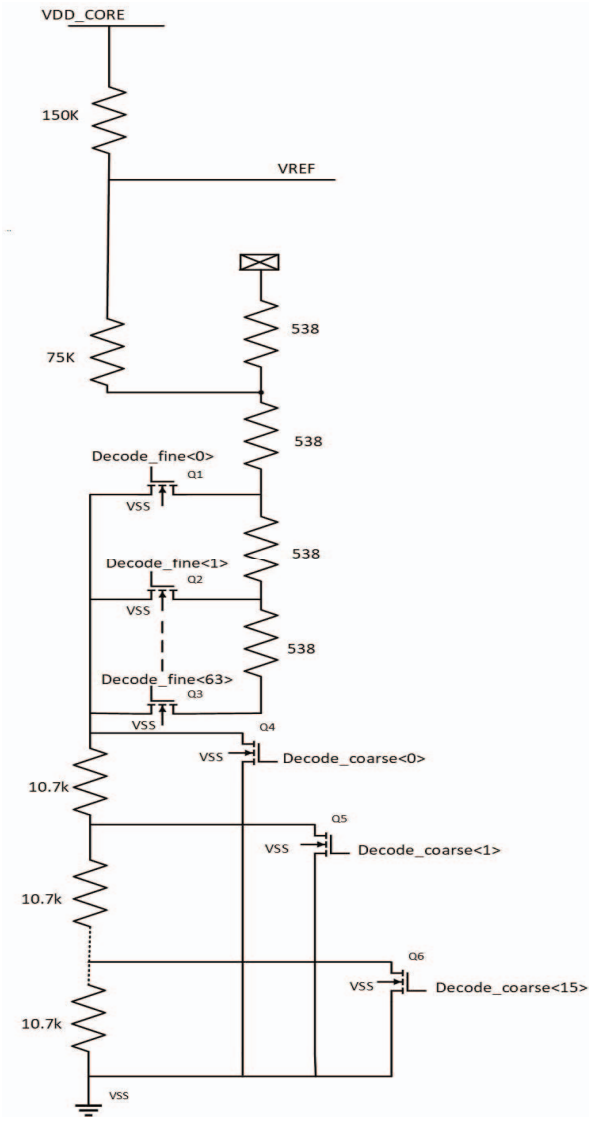


Fig.4. Example of a Resistance Ladder to generate the reference voltage

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TABLE I. Worst-case Leakage Current

Supply Voltage	MOS Corner	Resistor Corner	Layout	$I_{leakage}$		
				–40 °C	27 °C	125 °C
0.825 V	FF	FF	Initial	0.3 nA	5.2 nA	280 nA
			Improved	0.2 nA	2.6 nA	57 nA

TABLE II. Worst-case Reference Voltage

Supply Voltage	MOS Corner	Resistor Corner	Layout	V_{ref}		
				–40 °C	27 °C	125 °C
0.825 V	FF	FF	Initial	0.540 V	0.540 V	0.527 V
			Improved	0.541V	0.541 V	0.540 V

III. Overcoming Parasitics in Ring Oscillator

Given the compact, three-dimensional nature of the finFET structure and surrounding interconnect, parasitics—especially resistance—are increasing at a rapid rate. See Fig. 5. High parasitic resistance leads to supply drop, which significantly degrades FET performance, especially at the low supply voltage commonplace in finFET nodes. In addition, high parasitic capacitance due to strong gate coupling to diffusion contacts and overlying metal leads to many design iterations to reconcile the gap between initial schematic-based (pre-layout) and layout-based (post-layout) targets [7]. Consider the five-stage ring oscillator in Fig. 6, designed for 7.0 GHz. The oscillator is current-starved with a maximum amplitude (V_{ring}) of 0.825 V to avoid electrical overstress. A current of 2 mA was forced from an ideal source. Table III shows the gap between pre- and initial post-layout simulations. In this comparison, some estimated parasitics were already included in the schematic-based simulation. So, to achieve the 7.0 GHz frequency in post-layout, higher current has to be forced, resulting in V_{ring} exceeding the reliability limit.

To achieve the 7.0-GHz target, five major modifications were made to the layout. First, the two inverters highlighted in Fig.6 are placed together with a shared active area. With this continuous active area layout style layout, device mobility is maximized and constant for optimum matching and performance. The length of oxide definition (LOD) effect saturates after a sufficient number of dummy devices are appended to the edges of the continuous active area [7]. Secondly, minimum-length devices with slightly larger source/drain areas are used for reduced contact resistance and improved drive current.

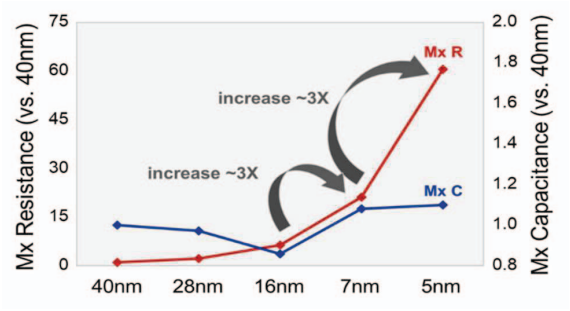


Fig. 5. Interconnect parasitics with technology scaling [5].

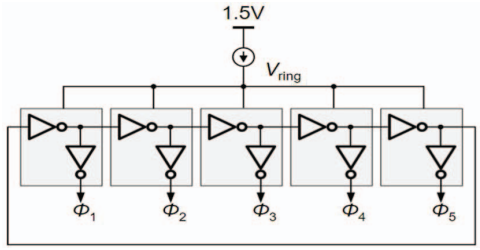


Fig. 6. Five-stage inverter-based ring oscillator.

TABLE III. Pre- and Post-layout Simulation Results

Design Criterion	*Pre-layout Simulation*	*Post-layout Simulation*
Current	2 mA	2 mA
Frequency	7.13 GHz	5.85 GHz
V_{ring}	0.666 V	0.804 V

Third, the usage of the Metal-0 (M0) metal layer over the MOSFET gates is minimized to reduce severe capacitance. M0 was mainly used for layer promotion to higher metal layers that exhibit far fewer parasitics. This is done by reducing the number of M0 routing tracks to 3-track M0, exchanging routing density for reduced parasitics. A wider M0 is used to further reduce resistance. A comparison is shown in Fig. 7. With the 3-track M0, V_{ring} is now reduced to 0.775 V and the frequency increased to 6.37 GHz, an almost 9% improvement. Fourth, the overhang of the diffusion contacts beyond the active area is minimized to reduce contact-to-gate capacitance [11]. The aspect ratio and floorplan of the oscillator's five stages are reworked to shorten inter-stage routes, as shown in Fig. 8. This helps in reducing the parasitic RC on the five phases as well as helps in matching their route length. A higher and wider metal layer is used for inter-stage routing, which helps in better parasitic resistance. The performance of several layout iterations is summarized in Fig. 9. With all of the above layout improvements, the ring oscillator can now run at 7.0 GHz and meet design specifications. See Table IV.

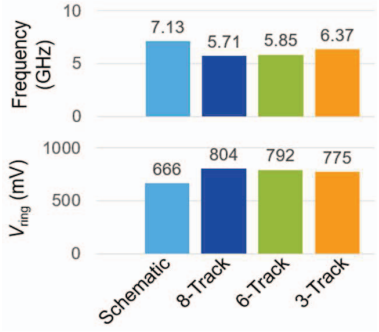


Fig. 7. Oscillator performance vs. number of M0 tracks.

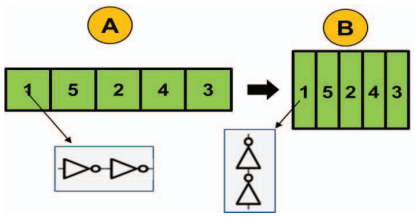


Fig. 8. Aspect ratio optimization to reduce parasitic.

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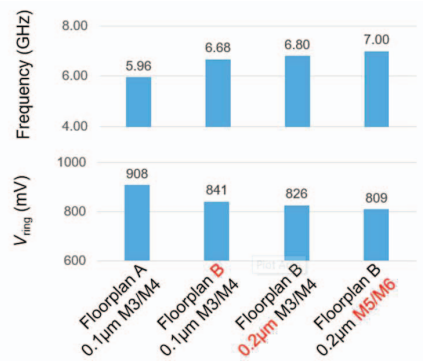


Fig.9. Comparison with floorplan A, B and with the use of higher and wider metal layer.

TABLE IV. Alignment between Pre- and Post-layout Simulations

Simulation Comparison			
Design Criterion	***Schematic Simulation***	***Initial Layout Simulation***	***Optimized Layout Simulation***
Current	2mA	2mA	**2mA**
Frequency	7.13GHz	5.85GHz	**7.01GHz**
Vring	0.666V	0.804V	**0.809V**

IV. Overcoming Gate-Induced Drain/Source Leakage

When a MOSFET is biased in the off state and voltage at the drain/source is high, significant drain to body/source to body leakage current can be observed (Fig. 11). GIDL/GISL currents result from tunneling of valence band electrons to the conduction band at the overlap region between gate and drain/source where strong vertical and longitudinal fields are present(Fig. 10). Mitigating this effect in 5nm CMOS is difficult. GIDL/GISL current increases on lowering the gate voltage because of the higher voltage difference between the gate and drain/source. This makes conventional leakage reduction techniques, e.g., negative charge pumps, much less effective to reduce GIDL.

The setup shown in Fig. 12 simulates 5nm device leakage. NMOS and PMOS are biased in the off state, a common-mode voltage (V_{cm}) is connected to the drain, and a voltage (Vx) connected to the source is swept from zero to supply.

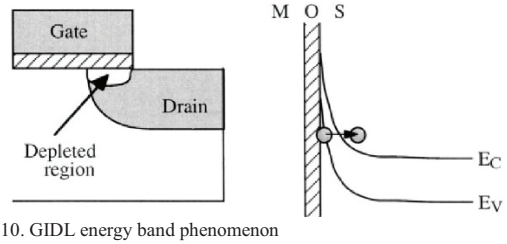


Fig. 10. GIDL energy band phenomenon

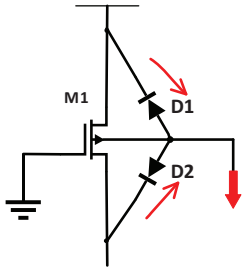


Fig. 11. Gate-induced drain leakage phenomenon.

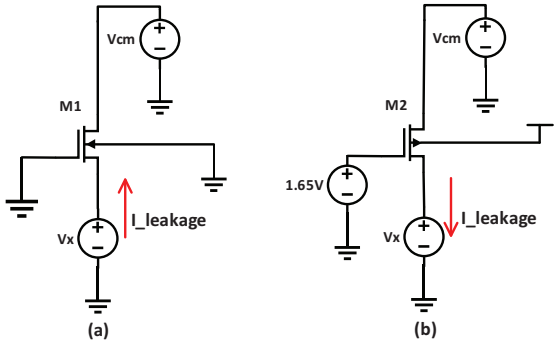


Fig. 12. Setup for (a) leakage in NMOS and (b) leakage in PMOS.

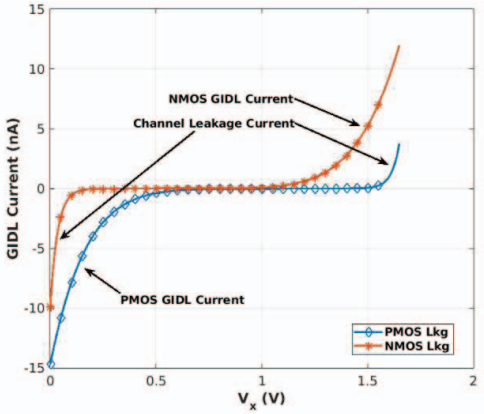


Fig. 13. NMOS and PMOS leakage in 5nm.

In the NMOS and PMOS leakage characteristics shown in Fig. 13, GIDL current increases exponentially for NMOS above 1.2 V and dominates below 0.4 V for PMOS. Controlling the device off-state leakage is critical for the performance of many mixed-signal blocks, including SAR ADC.

There are two phases in the SAR ADC operation: sampling and conversion. Fig. 14 shows the capacitor configuration in the SAR ADC. During the sampling phase, the input voltage (V_{in}) is sampled onto a sampling capacitor (C_s) by closing the input and top plate switches. During the conversion phase, the input switch and top plate switches are open. The charge stored in the sampling phase is redistributed among the capacitors in the DAC by opening and closing switches sw1/sw2.

The top plate of the capacitors is undriven in the conversion phase and any charge leakage through the top plate switch

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will change the value of the charge stored on the capacitor, resulting in an error. Due to the nonlinear nature of GIDL current, the error will be nonlinear and contribute to the DAC INL. The issue will be more pronounced for Synchronous SAR ADCs working on a slow clock. If the top plate switch is implemented with NMOS, the error will be highest when the top plate voltage is full-scale and negligible below 75 % of the full-scale input. If the switch is implemented with PMOS, the error will be highest when the top-plate voltage is zero and negligible above 400 mV. If the switch is implemented using a CMOS transmission gate, then the error would be maximum near zero and full-scale inputs and negligible near the mid-scale.

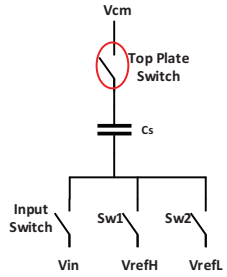


Fig. 14. Typical capacitor configuration in SAR ADC.

In 5nm, GIDL becomes significant for NMOS (biased in off state) at a drain voltage greater than 1.2 V and increases exponentially with increasing drain voltage. We can reduce the leakage by disallowing the voltage between the gate and drain/source to exceed 1.2 V. In Fig. 15, we have proposed a scheme to eliminate GIDL current during the conversion phase of a SAR ADC. By sensing whenever node A exceeds 1.2V, we can increase the gate potential by around 400 mV to lower the gate-to-drain voltage and thereby reduce the GIDL current. The Sense block is an amplifier followed by a Schmitt trigger for some built-in hysteresis. The sense block (S) output feeds a digital control block that selects the gate bias for the switch (M1) depending on the ADC mode and potential at node A. When the ADC is in the sampling phase, the switch is turned ON to apply common-mode voltage to the capacitor top plate. In the conversion phase, if the potential at node A is less than 1.2 V, S will be zero and the switch will be turned off with zero volts applied to the gate terminal. When the potential at node A exceeds 1.2 V, then the switch is turned off by applying around 400 mV at its gate. This will not allow the gate-to-drain voltage to go beyond 1.2 V and thereby limit the impact of GIDL. A similar scheme can be used for the PMOS switch. In Fig. 16, we have shown that when the proposed GIDL solution is enabled (for an NMOS switch), leakage is dramatically reduced when the top-plate voltage exceeds 1.2 V. Fig. 17 demonstrates the impact of the proposed GIDL solution on the error in sampled voltage. When the GIDL solution is enabled, there is no change in the top plate voltage during conversion.

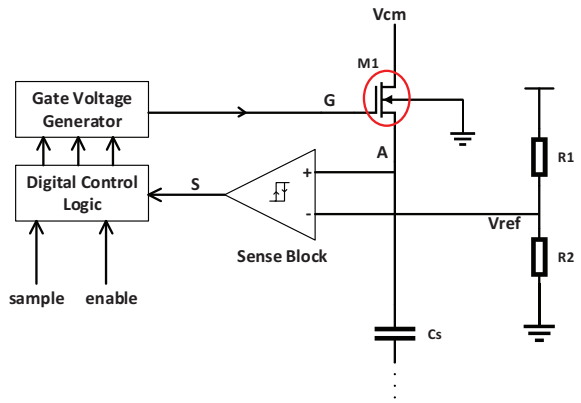


Fig. 15. Proposed solution to reduce GIDL in SAR ADC.

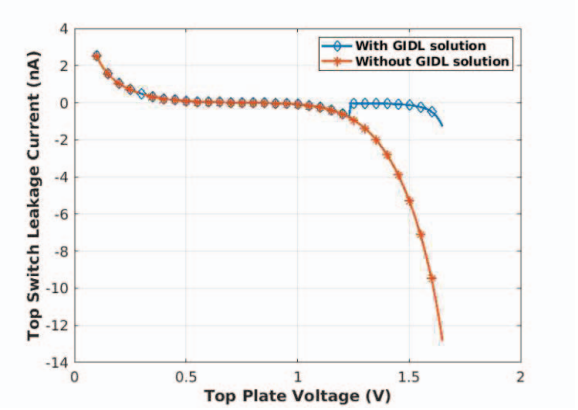


Fig. 16. NMOS Leakage with and without GIDL solution.

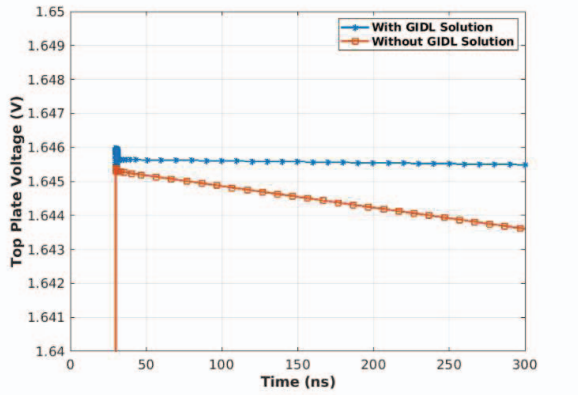


Fig. 17. Top Plate Voltage droop in conversion phase with and without GIDL Solution

V. Conclusions

This paper presented some key challenges in analog/mixed-signal designs as we scale to 5nm CMOS technology. This paper discussed how to reduce leakage currents in dummy transistors, interconnect parasitics to maximize ring oscillator performance and GIDL. These techniques were effective in overcoming issues imposed by the technology in order to meet the required design performance.

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REFERENCES

[1] M. Ryu, Y. Kang, and Y. Kim, "Transistor layout optimization for leakage saving," in *Int. SoC Design Conf.*, 2013, pp. 253–254, doi: 10.1109/ISOCC.2013.6864020.

[2] J. -H. Yu *et al*., "An inverter layout technique for propagation delay minimization," in *Int. Symp. Consumer Electronics*, 2015, pp. 1–2, doi: 10.1109/ISCE.2015.7177790.

[3] M. Madhusudan *et al*., "Analog layout generation using optimized primitives," in *Design, Automation & Test in Europe Conf. & Exhibition*, 2021, pp. 1234–1239, doi: 10.23919/DATE51398.2021.9474010.

[4] B. Jiang and T. Xia, "Model analysis of multi-finger MOSFET layout in ring oscillator design," in *IEEE Int. Symp. Quality Electronic Design*, 2011, pp. 1–6, doi: 10.1109/ISQED.2011.5770749.

[5] C. Hou, "A smart design paradigm for smart chips," in *IEEE Int. Solid-State Circuits Conf*, 2017, pp. 8–13, doi: 10.1109/ISSCC.2017.7870240.

[6] S. Yang *et al*., "High-performance mobile SoC design and technology co-optimization to mitigate high-K metal gate process variations," in *IEEE Symp. VLSI Technology*, 2014, pp. 1–2, doi: 10.1109/VLSIT.2014.6894381

[7] A. L. S. Loke *et al*., "Analog/mixed-signal design challenges in 7-nm CMOS and beyond," in *IEEE Custom Integrated Circuits Conf.*, 2019, pp. 1–8, doi: 10.1109/CICC.2019.8780346.

[8] S. -Y. Wu *et al*., "A 16nm FinFET CMOS technology for mobile SoC and computing applications," in *IEEE Int. Electron Devices Meeting*, 2013, pp. 9.1.1–9.1.4, doi: 10.1109/IEDM.2013.6724591.

[9] H.-J. Cho *et al*., "Si FinFET based 10nm technology with multi Vt gate stack for low power and high performance applications," in *IEEE Symp. VLSI Technology*, 2016, pp. 1–2, doi: 10.1109/VLSIT.2016.7573359.

[10] Y.-K. Choi, T.-J. King, and Chenming Hu, "A spacer patterning technology for nanoscale CMOS," *IEEE Trans. Electron Devices*, vol. 49, no. 3, pp. 436–441, Mar. 2002, doi: 10.1109/16.987114.

[11] C. Auth *et al*., "A 10nm high performance and low-power CMOS technology featuring 3rd generation FinFET transistors, self-aligned quad patterning, contact over active gate and cobalt local interconnects," in *IEEE Int. Electron Devices Meeting*, 2017, pp. 29.1.1–29.1.4, doi: 10.1109/IEDM.2017.8268472.

[12] Y. Woo *et al*., "Design and process technology co-optimization with SADP BEOL," in sub-10nm SRAM bitcell," in *IEEE Int. Electron Devices Meeting*, 2015, pp. 11.2.1–11.2.4, doi: 10.1109/IEDM.2015.7409674.

[13] L. Chang *et al*., "Gate length scaling and threshold voltage control of double-gate MOSFETs," in *IEEE Int. Electron Devices Meeting*, 2000, pp. 719–722, doi: 10.1109/IEDM.2000.904419.

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Large Language Model (LLM) for Standard Cell Layout Design Optimization

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***Abstract*—Standard cells are essential components of modern digital circuit designs. With process technologies advancing toward 2*nm*, more routability issues have arisen due to the decreasing number of routing tracks, increasing number and complexity of design rules, and strict patterning rules. The state-of-the-art standard cell design automation framework is able to automatically design standard cell layouts in advanced nodes, but it is still struggling to generate highly competitive Performance-Power-Area (PPA) and routable cell layouts for complex sequential cell designs. Consequently, a novel and efficient methodology incorporating the expertise of experienced human designers to incrementally optimize the PPA of cell layouts is highly necessary and essential.**

High-quality device clustering, with consideration of netlist topology, diffusion sharing/break and routability in the layouts, can reduce complexity and assist in finding highly competitive PPA, and routable layouts faster. In this paper, we leverage the natural language and reasoning ability of Large Language Model (LLM) to generate high-quality cluster constraints incrementally to optimize the cell layout PPA and debug the routability with ReAct prompting. On a benchmark of sequential standard cells in 2*nm*, we demonstrate that the proposed method not only achieves up to 19.4% smaller cell area, but also generates 23.5% more LVS/DRC clean cell layouts than previous work. In summary, the proposed method not only successfully reduces cell area by 4.65% on average, but also is able to fix routability in the cell layout designs.

I. Introduction

Standard cells are essential components of modern digital circuit designs. As process technologies relentlessly advance toward 2*nm*, designing a cell with competitive Performance-Power-Area (PPA) while considering routability becomes increasingly challenging due to the decreasing number of routing tracks, increasing complexity of design rules, and strict patterning rules. The state-of-the-art standard cell design automation framework is capable of automatically designing standard cell layouts in advanced nodes, but it still struggles to generate highly competitive PPA and routable cell layouts for complex sequential cell designs. As a result, a novel and efficient methodology, leveraging the expertise of experienced human designers, to optimize the PPA and routability of cell layouts incrementally, has emerged as a critical need.

Recently, automated standard cell synthesis tools such as NVCell [1] and BonnCell [2], have been shown to generate high quality cell layouts on advanced technology nodes. Due to routability issues, one of the key challenges is that the generated placement for any given cell could be unroutable or unable to be routed without DRC errors. NVCell2 [3] develops a lattice graph routability model and successfully improves the routability in the advanced technology nodes. However, its performance is not scale to hundreds of transistors because the model inference needs to be performed for every action in the simulated annealing-based placement algorithm [1] and the cell-level metrics (i.e., cell width (CW) and total wirelength (TWL)) are compromised for routability. Ho *et al.* [4] proposed a transformer model-based cluster approach to generate high-quality device cluster constraints, which considers diffusion sharing/break, routability, and DRCs of routing metals in the layout of different technology nodes, and achieved better PPA, routability, and performance than [3] in the advanced nodes. However, selecting a good set of LVS/DRC clean layouts to train the transformer cluster model for optimizing PPA and routability of complex sequential cells together is quite challenging. This is because cells with routability issues typically have a larger cell width to reduce transistor pin density, while cells with a more compact layout could exacerbate routability issues. Additionally, there is a limited amount of LVS/DRC clean layouts available for training the transformer cluster model in the early development stage of cell library in a new technology node.

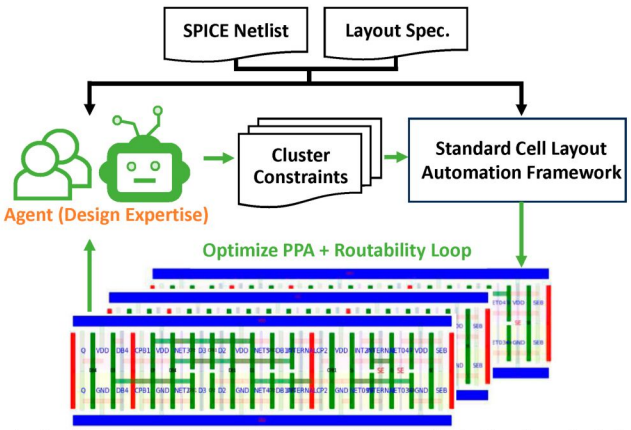


Fig. 1: An illustration of the PPA and routability optimization loop includes an agent with design expertise, adjusted cluster constraints, and a standard cell layout automation framework (i.e., NVCell).

An agent with designers' expertise can adjust and fine-tune device clustering constraints incrementally based on the netlist and the layout of the previous iteration to efficiently optimize PPA and routability together as shown in Figure 1. Lately, Large Language Models (LLMs) have shown great promise across various tasks in language understanding and interactive decision-making, incorporating reasoning and actions. In this paper, we leverage the natural language and reasoning ability of LLMs to adjust the device clustering constraints incrementally,

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optimizing cell layout PPA and routability with guidance from designers' expertise and ReAct [5] prompting techniques. Our main contributions are as follows.

- We are the first to explore LLM for optimization in Electronic Design Automation (EDA) on an industrial-level benchmark. We propose a novel, and efficient LLM for standard cell layout design optimization methodology to generate high-quality cluster constraints to optimize the cell layout PPA and debug the routability with guidance of designers' expertise and ReAct [5] prompting techniques on an industrial technology node. The proposed methodology can improve the cell-level PPA and generate the device clusters incrementally considering netlist, previous cluster constraints, routability, and physical layout, simultaneously.
- We conduct holistic assessments and studies on the capabilities and domain knowledge of existing LLM on SPICE netlist language, cluster design constraint format, and physical layout description. Then, we automate the domain knowledge extraction with guidelines from designers' expertise for optimizing the PPA and routability together.
- The proposed novel LLM for standard cell layout design optimization methodology achieves up to 19.4% smaller cell area, and generates 23.5% more LVS/DRC clean cell layouts than a state-of-the-art baseline on a benchmark of sequential standard cells in industrial 2*nm* technology node.

The remaining sections are organized as follows: Section II demonstrates the study and assessment of existing LLM on understanding the netlist and standard cell layout domain knowledge. Section III describes our novel LLM for standard cell layout design optimization methodology. Section IV presents our main experiment. Section V concludes the paper.

II. Standard Cell Layout Design Domain Knowledge

We conduct assessments and studies on the capabilities and domain knowledge of existing LLMs of SPICE language format, device cluster constraints, and physical layout of standard cells.

A. SPICE Netlist Language

To assess the domain knowledge of LLM on SPICE netlist language, we input the technology independent device (i.e., MOSFET) description in SPICE language format and let LLM explain the MOSFET information. The technology independent description of a MOSFET includes name, terminal connections, and the type of MOSFET. Figure 2(a) shows that LLM can understand technology independent MOSFET description in SPICE language format and identify the net connection information at the drain, gate, and source terminals[1].

B. Cluster Constraint

We study the ability of existing LLM on understanding the cluster design constraint format including multiple devices and clusters. We show one of the example studies in Figure 2(b). We task the LLM with summarizing the provided clusters and the devices associated with them for subsequent reasoning tasks. The study's findings indicate that the existing LLM can accurately identify the total number of clusters and the information regarding devices within these clusters.

[1]The [MOSFET format] is "A MOSFET can be described in SPICE format as MOSFET_NAME d:DRAIN g:GATE s:SOURCE MOSFET_TYPE".

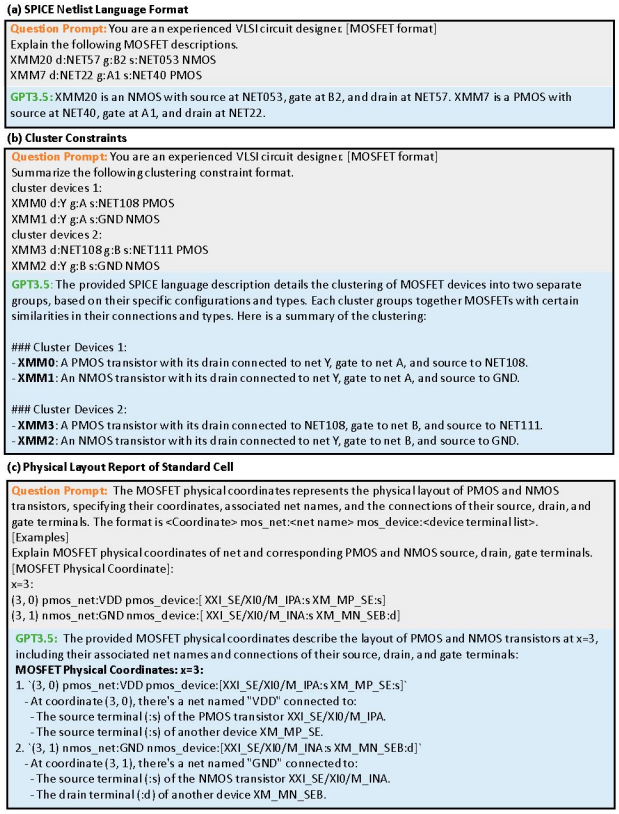


Fig. 2: Standard cell layout design domain knowledge assessment of existing LLM on: (a) SPICE language, (b) cluster constraint, and (c) physical layout.

C. Standard Cell Layout

The standard cell layout includes the placed device locations and the net connection of device terminals. We use coordinates and the corresponding device and its terminals to represent the standard cell layout, as shown in Figure 2(c). We study the capability of LLM in understanding the location of placed devices and transistor terminal connections for the standard cell layout design optimization task. In one of the study examples in Figure 2(c), the LLM successfully explains the placed MOSFETs and their terminals at each coordinate.

In summary, the existing LLM (i.e., GPT3.5) has the capability to understand the netlist topology through SPICE language, device cluster constraint, and device placement, and the terminal connections in standard cell layout description for generating better device cluster constraints for PPA and routability debugging.

III. LLM for Standard Cell Layout Design Optimization

We introduce the details of LLM for standard cell layout design optimization, which leverages the natural language and reasoning ability of LLM to generate high-quality cluster constraints to optimize the cell-level PPA and debug routability issues. The overall flow is outlined in Section III-A, and Figure 3. Section III-B introduces the developed netlist tools, which provide accurate sub-circuit retrieval from the netlist and cluster evaluation, for assisting LLM on layout design optimization task. Finally, we discuss the application of ReAct [5] in Section III-C.

A. Flow Overview

The proposed LLM for standard cell layout design optimization comprises the following components: knowledge extraction

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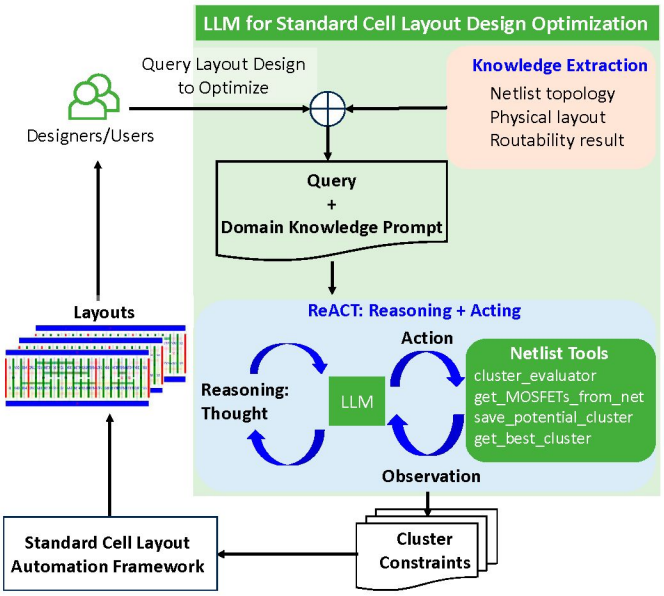


Fig. 3: Overview of LLM for Standard cell layout design optimization flow

to initiate queries and provide domain knowledge prompts, netlist tools to assist the LLM in generating valid cluster constraints, and reasoning and action in ReAct [5] for exploring high-quality cluster candidates for the PPA/routability of layout design.

Firstly, designers input the initial layout and its corresponding cluster constraints. The knowledge extraction is used to create the domain knowledge prompt with netlist topology in technology independent descriptions of MOSFETs, initial cluster constraints, standard cell layout, and guidance from designers' expertise as described in Section II. Then, the ReAct prompting method allows LLM to perform dynamic reasoning to create and adjust plans for acting (reason to act), while also interacting with the netlist tools (i.e., grouping MOSFET, evaluating clusters, etc.) to incorporate additional information into reasoning (act to reason). These cluster constraints can be fed into NVCell [3] to generate layouts. Designers can repeat this process until the PPA meets the requirements without rouability issues.

B. Netlist Tools

We have developed a set of netlist tools to assist LLM in generating valid cluster constraints and correctly retrieving sub-circuits in the ReAct reasoning and action loop. The netlist tools include a cluster evaluator, a function to retrieve group devices from nets, a mechanism to save potential clusters, and a function to obtain the best cluster result.

Cluster evaluator: This tool is used to evaluate the quality of the generated cluster result using simple cluster score to capture the potential diffusion sharing and common gate in the layout. Here, we use the simple cluster score for evaluation in ReAct because the turnaround time is too long to launch layout generation to collect accurate cell layout metrics (i.e., CW, TWL, etc.). The simple cluster score is calculated in Equation (1). A larger score means the devices can potentially be placed with more common diffusion sharing and common gate inside each cluster.

$$cluster_score = \sum_{c \in \mathbf{C}} \left(\frac{\sum_{n \in \mathbf{N_c^d}} \lfloor \frac{P_n}{2} \rfloor + \lfloor \frac{N_n}{2} \rfloor}{T_c} + \frac{\sum_{n \in \mathbf{N_c^g}} \min(P_n, N_n)}{T_c} \right) \quad (1)$$

Where c is a cluster in a set of clusters, $\mathbf{C}$. $\mathbf{N_c^d}$ and $\mathbf{N_c^g}$ represent the set of nets at diffusion (i.e., source or drain terminals) and gate in cluster c. P_n and N_n represent the number of net n at PMOS and NMOS terminals, respectively. T_c is the number of transistors in cluster c. For potential diffusion sharing in a cluster, we calculate the number of same net pairs at diffusion terminals for PMOS and NMOS, respectively. The minimum number of the same gate net of PMOS and NMOS is the potential number of common gates in a cluster.



Fig. 4: An example of ReAct steps of Thought-Action-Observation for standard cell layout design optimization.

Get group devices from nets: This netlist tool returns the group of transistors from an arbitrary number of nets in the netlist. LLM can use this tool to search and explore potential device clusters.

Save potential cluster: This tool returns the current clusters and cluster score after LLM inputs a new potential cluster. The duplicated devices in different clusters are fixed based on the number of shared nets of these duplicated devices in each cluster.

Get best cluster result: This tool returns the cluster result with the best simple cluster score (i.e., Equation (1)). It helps LLM revert back or restart the search from the previous best cluster result when it is stuck in the searching potential cluster phase.

C. ReAct: Reason + Act Loop

We enable LLMs to function as autonomous circuit design agent for reasoning and acting with the netlist tools through **ReAct** prompting mechanism [5]. In ReAct, the LLM initiates the generation of subsequent steps with Thought, Action, and Observation components. The action is querying one of the netlist tools described in Section III-B. The output of the queried netlist tool from action becomes the observation in the prompt. The agent continues the reasoning and acting traces until selecting the "Final Answer" action. Figure 4 shows an example of optimizing the cell area of a standard cell. Here, the agent starts with querying the group of devices connected to NET027 to explore

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good clusters incrementally to reduce the diffusion break for area reduction since NET027 is one of the high connection nets in the netlist topology and abutted to the diffusion break dummy device in the physical layout. Finally, the agent successfully generates high-quality cluster result through reasoning and leveraging the netlist tools traces in ReAct.

IV. Experimental Results

Our work is implemented with Python and LangChain [7]. We conduct all experiments with gpt-3.5-turbo-16k-0613 as the LLM through OpenAI APIs [8]. We set the sampling temperature of LLM to 0.1. For ReAct prompting, we restrict the LLM to a maximum of 15 iterations of Thought-Action-Observation.

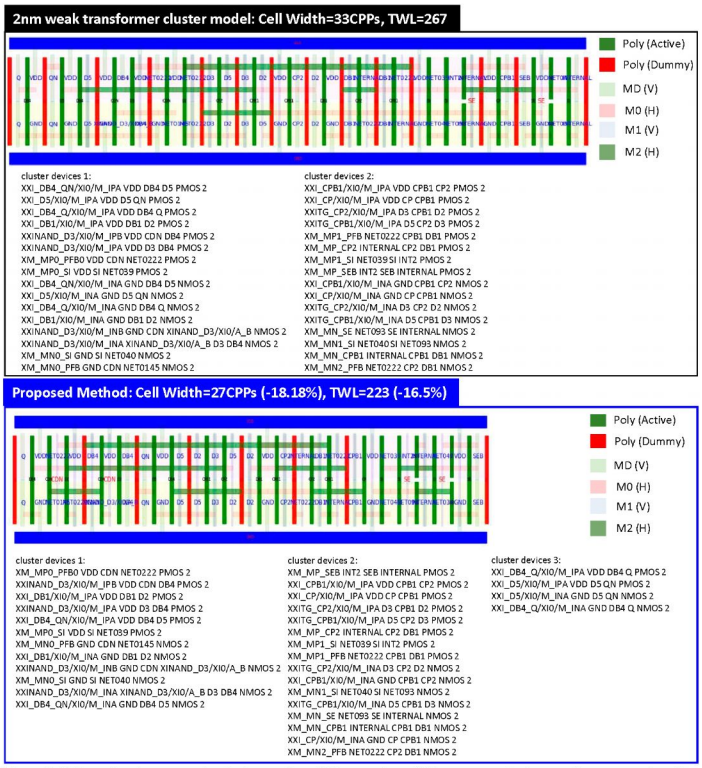


Fig. 5: The cluster constraints and cell layouts of ***2nm* weak transformer cluster model** [4], and proposed method of Seq3 cell. The proposed method successfully reduce the CW, and TWL by 18.18%, and 16.48%, respectively.

TABLE I: CW (CPPs) table of the 17 complex sequential cells in 2*nm* of transformer cluster model [4], simulated annealing (SA), and the proposed method. The SA algorithm is implemented base on [6]. #Devs=number of transistors. Opt.(a)/(b)=Optimize layouts from (a)/(b). X=unroutable. V=Fix routability. Success(%)=the percentage of LVS/DRC clean cell layouts.

Cell	#Devs	Transformer Cluster [4]		SA		Proposed Method			
								Impr. (%)	
		(a) 2nm weak	(b) 5nm	Opt. (a)	Opt. (b)	Opt. (a)	Opt. (b)	Over (a)	Over (b)
Seq1	40	26	31	X	31	25	25	3.85	19.35
Seq2	60	X	41	X	X	39	39	V	4.88
Seq3	40	33	26	27	26	27	26	18.18	0.00
Seq4	38	27	25	X	25	25	25	7.41	0.00
Seq5	36	23	22	26	22	22	22	4.35	0.00
Seq6	36	22	25	22	22	22	22	0.00	12.00
Seq7	34	20	20	20	20	20	20	0.00	0.00
Seq8	38	25	25	25	25	24	25	4.00	0.00
Seq9	32	19	19	19	19	19	19	0.00	0.00
Seq10	34	20	21	20	20	20	20	0.00	4.76
Seq11	40	26	26	25	28	25	25	3.85	3.85
Seq12	38	24	27	26	28	24	24	0.00	11.11
Seq13	56	X	X	X	X	41	40	V	V
Seq14	44	X	36	X	X	34	34	V	5.56
Seq15	42	X	32	X	X	31	31	V	3.13
Seq16	56	42	40	X	40	35	35	16.67	12.50
Seq17	42	25	25	25	25	25	25	0.00	0.00
Success (%)		76.50	94.10	58.80	76.50	100	100	-	-

We conduct extensive studies on the cell area and routability using 17 complex sequential standard cells in industrial 2*nm* technology node. Due to a lack of cell layout data in the early stage development of cell library in 2*nm*, we trained (a) ***2nm* weak transformer cluster model** [4] using 124 LVS/DRC clean cell layouts, and (b) ***5nm* transformer cluster model** [4] using 512 LVS/DRC clean cell layouts as the baselines. We implemented a simulated annealing algorithm (SA), which is based on the modified Lam annealing scheduler [6] that required no hyper parameter tuning, for comparing the efficiency of the proposed method. In SA, we sample arbitrary 1 to k nets considering the weights of nets and querying a group of devices from **Get group devices from nets** tool for each *action*. The unrouted nets, and nets abutted to diffusion break dummy device have higher weights. Then, this queried group of devices will be accepted or rejected to be saved using **Save potential cluster** tool based on the temperature and the delta simple cluster score. Finally, we apply the proposed method, and SA to optimize the cell area and routability from the initial cluster constraints and cell layouts of (a), and (b) on the selected unseen 17 complex sequential cells. For the proposed method, we launch 10 optimization runs for each cell and select valid cluster results to generate cell layout using NVCell [3].

Table I shows the CW (CPPs) of the selected 17 sequential cells of transformer cluster method [4], SA, and the proposed method. SA fails to optimize the cell area and fix the routability because the simple cluster score can not fully capture diffusion sharing, and routability in the layout. It requires more holistic and efficient methodology for cluster exploration than a naive net sampling-based approach. Compared to (a), the proposed method can not only reduce up to 18.18% of CW, but also improve the success rate from 76.5% to 100%. In addition, the proposed method reduces up to 19.35% of CW, and improves the success rate from 94.10% to 100% when optimizing the initial cell layout from (b). Figure 5 shows the layouts and generated cluster constraints of (a), and using proposed method to optimize from (a) of Seq3 cell. The proposed method achieves 18.18% reduction in CW, and 16.48% reduction in TWL.

In summary, excluding the fix routability cells, the proposed method achieves 4.48%, and 4.82% reduction in cell area on average over (a), and (b). We successfully demonstrate that the proposed method incorporates holistic information from the netlist and layout of complex sequential cells to conduct efficient cluster exploration with a simple cluster score for the standard cell layout design optimization task.

V. Conclusion

We propose a novel, efficient, and the first LLM for standard cell layout design optimization methodology to generate high-quality cluster constraints to optimize the cell layout PPA and debug the routability with the guidance of designers' expertise and ReAct [5] prompting techniques. We have demonstrated that the proposed method achieves up to 19.4% smaller cell area, and generates 100% LVS/DRC clean cell layouts on the selected complex sequential cell benchmark in 2*nm*. This research not only provides a novel autonomous LLM agent for standard cell layout design optimization and debugging but also introduces the new application of using LLM assistance for optimization in the EDA field for further exploration.

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REFERENCES

[1] Haoxing Ren and Matthew Fojtik. Nvcell: Standard cell layout in advanced technology nodes with reinforcement learning. In *2021 58th ACM/IEEE Design Automation Conference (DAC)*, pages 1291–1294. IEEE, 2021.

[2] Pascal Van Cleeff, Stefan Hougardy, Jannik Silvanus, and Tobias Werner. Bonncell: Automatic cell layout in the 7-nm era. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 39(10):2872–2885, 2019.

[3] Chia-Tung Ho, Alvin Ho, Matthew Fojtik, Minsoo Kim, Shang Wei, Yaguang Li, Brucek Khailany, and Haoxing Ren. Nvcell 2: Routability-driven standard cell layout in advanced nodes with lattice graph routability model. In *Proceedings of the 2023 International Symposium on Physical Design*, pages 44–52, 2023.

[4] Chia-Tung Ho, Ajay Chandna, David Guan, Alvin Ho, Minsoo Kim, Yaguang Li, and Haoxing Ren. Novel transformer model based clustering method for standard cell design automation. In *Proceedings of the 2024 International Symposium on Physical Design*, pages 195–203, 2024.

[5] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*, 2022.

[6] Vincent A Cicirello. On the design of an adaptive simulated annealing algorithm. In *Proceedings of the international conference on principles and practice of constraint programming first workshop on autonomous search*, 2007.

[7] Harrison Chase. LangChain, October 2022.

[8] OpenAI. Openai models api. 2023.

VI. APPENDIX

We present examples of netlist topology, physical layout, and routability report prompts from proposed knowledge extraction component as mentioned in Fig. 4. We describe the netlist topology and physical layout prompts of the simple cell example (i.e., OA333X1) in Fig. 4. For the routability report prompt, we select the Seq13 in Table I for demonstration, as there are no unrouted nets in the OA333X1 layout depicted in Fig. 4.

A. Netlist topology prompt

The netlist topology prompt consists of MOSFET connection and description, as well as previous cluster constraints, as shown in Fig. 6. In the MOSFET connection and description, each MOSFET is described using the technology-independent device description format introduced in Section II. For the previous cluster constraints, we present them in JSON BLOB format with the action labeled as "Final Answer". At the end, we provide the simple cluster score and the number of clusters resulting from the previous cluster constraints.

With the netlist topology prompt information, the LLM agent can understand the netlist connections of each device, previous cluster constraints, and the simple cluster score of previous cluster constraints for the following reasoning and action for cell layout design optimization.

MOSFET Connection and Description

```
[Circuit Information begin]:
CELLNAME OA333X1
== MOSFET_NAME SOURCE GATE DRAIN MOSFET_TYPE ==
XMM20 s:Y g:NET027 d:VDD PMOS
XMM19 s:NET087 g:B2 d:NET058 PMOS
XMM23 s:VDD g:A3 d:VDD PMOS
XMM13 s:NET040 g:C3 d:VDD PMOS
XMM14 s:NET027 g:A1 d:NET084 PMOS
XMM16 s:NET086 g:C2 d:NET040 PMOS
XMM12 s:NET027 g:B1 d:NET087 PMOS
XMM11 s:NET058 g:B2 d:NET041 PMOS
XMM17 s:NET027 g:C1 d:NET086 PMOS
XMM3 s:NET038 g:A3 d:VDD PMOS
XMM10 s:NET041 g:B3 d:VDD PMOS
XMM2 s:NET084 g:A2 d:NET038 PMOS
XMM22 s:GND g:NET027 d:Y NMOS
XMM18 s:NET068 g:A3 d:NET068 NMOS
XMM21 s:NET068 g:B2 d:NET068 NMOS
XMM4 s:NET015 g:B1 d:NET068 NMOS
XMM1 s:GND g:C3 d:NET015 NMOS
XMM8 s:NET068 g:A2 d:NET027 NMOS
XMM9 s:NET068 g:A3 d:NET027 NMOS
XMM6 s:NET015 g:B3 d:NET068 NMOS
XMM7 s:NET068 g:A1 d:NET027 NMOS
XMM15 s:GND g:C1 d:NET015 NMOS
XMM5 s:NET015 g:B2 d:NET068 NMOS
XMM0 s:GND g:C2 d:NET015 NMOS
```

Previous Cluster Constraints

```
[Previous Cluster Result]:
AI: Action:
```
{"action": "Final Answer",
"action_input":"
cluster devices 1:
XMM20 s:Y g:NET027 d:VDD PMOS
XMM19 s:NET087 g:B2 d:NET058 PMOS
XMM14 s:NET027 g:A1 d:NET084 PMOS
XMM12 s:NET027 g:B1 d:NET087 PMOS
XMM11 s:NET058 g:B2 d:NET041 PMOS
XMM17 s:NET027 g:C1 d:NET086 PMOS
XMM22 s:GND g:NET027 d:Y NMOS
XMM21 s:NET068 g:B2 d:NET068 NMOS
XMM8 s:NET068 g:A2 d:NET027 NMOS
XMM9 s:NET068 g:A3 d:NET027 NMOS
XMM7 s:NET068 g:A1 d:NET027 NMOS
XMM5 s:NET015 g:B2 d:NET068 NMOS
"
}
```
Cluster score of shared source, drain, and gate terminals: 0.004166666666666721
Number of clusters: 1
```

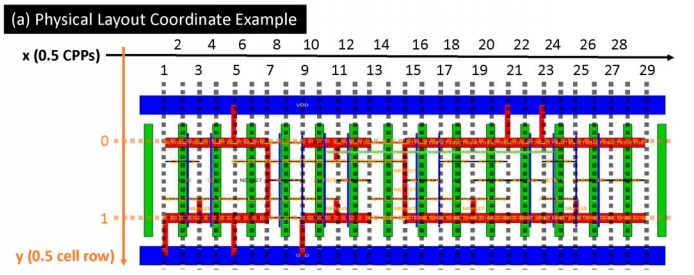
Fig. 6: An example of extracted netlist topology prompt of the simple cell example in Fig. 4.

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B. Physical layout prompt

We design the physical layout prompt with the placed device locations and net connection of device terminals for LLM to compile the netlist topology and layout together for ReAct steps of Thought-Action-Observation.

Fig. 7 shows the example of extracted physical layout prompt of OA333X1 layout. From the Fig. 7(a), the unit of x coordinate is the half contacted-poly-pitch (CPP), and the unit of y coordinate is half cell row. As a result, there are 29 columns and 2 rows in OA333x1 layout in the extracted coordinate based physical layout prompt in Fig. 7(b). For each coordinate, we show the net name, placed device, and the terminals (i.e., source, drain, gate) of the placed device. The net name, and device are dummy when there are no devices in the netlist being placed at the coordinate. Here, the column-based physical layout report format helps LLM identify the common gate and diffusion connections of PMOS and NMOS.



(b) Physical Layout Prompt

```
[OA333X1 MOSFET Physical Coordinate Report of Previous Cluster Result]:
[MOSFET Physical Coordinate]:
x=1:
(1, 0) pmos_net:NET086 pmos_device:[ XMM16:s]
(1, 1) nmos_net:GND nmos_device:[ XMM0:s]
x=2:
(2, 0) pmos_net:C2 pmos_device:[XMM16:g]
(2, 1) nmos_net:C2 nmos_device:[XMM0:g]
x=3:
(3, 0) pmos_net:NET040 pmos_device:[ XMM13:s XMM16:d]
(3, 1) nmos_net:NET015 nmos_device:[ XMM1:d XMM0:d]
x=4:
(4, 0) pmos_net:C3 pmos_device:[XMM13:g]
(4, 1) nmos_net:C3 nmos_device:[XMM1:g]
x=5:
(5, 0) pmos_net:VDD pmos_device:[ XMM20:d XMM13:d]
(5, 1) nmos_net:GND nmos_device:[ XMM22:s XMM1:s]
x=6:
(6, 0) pmos_net:NET027 pmos_device:[XMM20:g]
(6, 1) nmos_net:NET027 nmos_device:[XMM22:g]
x=7:
(7, 0) pmos_net:Y pmos_device:[ XMM20:s]
(7, 1) nmos_net:Y nmos_device:[ XMM22:d]
x=8:
(8, 0) pmos_net:dummy pmos_device:[dummy:g]
(8, 1) nmos_net:dummy nmos_device:[dummy:g]
x=9:
(9, 0) pmos_net:NET086 pmos_device:[ XMM17:d]
(9, 1) nmos_net:GND nmos_device:[ XMM15:s]
x=10:
(10, 0) pmos_net:C1 pmos_device:[XMM17:g]
(10, 1) nmos_net:C1 nmos_device:[XMM15:g]
x=11:
(11, 0) pmos_net:NET027 pmos_device:[ XMM12:s XMM17:s]
(11, 1) nmos_net:NET015 nmos_device:[ XMM4:s XMM15:d]
x=12:
(12, 0) pmos_net:B1 pmos_device:[XMM12:g]
(12, 1) nmos_net:B1 nmos_device:[XMM4:g]
x=13:
(13, 0) pmos_net:NET087 pmos_device:[ XMM19:s XMM12:d]
(13, 1) nmos_net:NET068 nmos_device:[ XMM21:d XMM4:d]
x=14:
(14, 0) pmos_net:B2 pmos_device:[XMM19:g]
(14, 1) nmos_net:B2 nmos_device:[XMM21:g]
x=15:
(15, 0) pmos_net:NET058 pmos_device:[ XMM19:d XMM11:s]
(15, 1) nmos_net:NET068 nmos_device:[ XMM21:s XMM5:d]
x=16:
(16, 0) pmos_net:B2 pmos_device:[XMM11:g]
(16, 1) nmos_net:B2 nmos_device:[XMM5:g]
x=17:
(17, 0) pmos_net:NET041 pmos_device:[ XMM11:d]
(17, 1) nmos_net:NET015 nmos_device:[ XMM5:s]
x=18:
(18, 0) pmos_net:dummy pmos_device:[dummy:g]
(18, 1) nmos_net:dummy nmos_device:[dummy:g]
x=19:
(19, 0) pmos_net:NET027 pmos_device:[ XMM14:s]
(19, 1) nmos_net:NET027 nmos_device:[ XMM7:d]
x=20:
(20, 0) pmos_net:A1 pmos_device:[XMM14:g]
(20, 1) nmos_net:A1 nmos_device:[XMM7:g]
x=21:
(21, 0) pmos_net:NET084 pmos_device:[ XMM14:d XMM2:s]
(21, 1) nmos_net:NET068 nmos_device:[ XMM8:s XMM7:s]
x=22:
(22, 0) pmos_net:A2 pmos_device:[XMM2:g]
(22, 1) nmos_net:A2 nmos_device:[XMM8:g]
x=23:
(23, 0) pmos_net:NET038 pmos_device:[ XMM3:s XMM2:d]
(23, 1) nmos_net:NET027 nmos_device:[ XMM8:d XMM9:d]
x=24:
(24, 0) pmos_net:A3 pmos_device:[XMM3:g]
(24, 1) nmos_net:A3 nmos_device:[XMM9:g]
x=25:
(25, 0) pmos_net:VDD pmos_device:[ XMM23:d XMM3:d]
(25, 1) nmos_net:NET068 nmos_device:[ XMM18:s XMM9:s]
x=26:
(26, 0) pmos_net:A3 pmos_device:[XMM23:g]
(26, 1) nmos_net:A3 nmos_device:[XMM18:g]
x=27:
(27, 0) pmos_net:VDD pmos_device:[ XMM23:s XMM10:d]
(27, 1) nmos_net:NET068 nmos_device:[ XMM18:d XMM6:d]
x=28:
(28, 0) pmos_net:B3 pmos_device:[XMM10:g]
(28, 1) nmos_net:B3 nmos_device:[XMM6:g]
x=29:
(29, 0) pmos_net:NET041 pmos_device:[ XMM10:s]
(29, 1) nmos_net:NET015 nmos_device:[ XMM6:s]
```

Fig. 7: An example of extracted physical layout prompt of the simple cell example in Fig. 4.

C. Routability report prompt

We select the Seq13 in Table I for describing the routability report prompt format since there are no unrouted nets in OA333X1 layout as shown in Fig. 8. In the routability report, we provide information about the unrouted nets, the corresponding pairs of x-coordinates of net terminals, and the placed devices inside the unrouted region. These placed devices within the unrouted region offer insight into routing congestion and required transistor pin access, assisting LLM in identifying potential good cluster constraints to improve routability. For example, if routing congestion or pin density is too high in an unrouted net region, leveraging common transistor terminal sharing across PMOS and NMOS, as well as diffusion sharing, can reduce pin density and routing resource usage by creating cluster constraints that consider the high connection nets or problematic nets of transistor pins in an unrouted net region.

```
[Route Evaluation]: Failed to route using Previous Cluster Result
[Failed to route net NET206]:
Failed to route net region: from x=6 to x=14
devices in the unrouted region:
XXI_SE/XI0/M_IPA VDD SE SEB PMOS
XM_MP_EB NET247 EB NET234 PMOS
XM_MP_E NET235 E NET234 PMOS
XM_MP_D5 VDD D5 NET235 PMOS
XXI_SE/XI0/M_INA GND SE SEB NMOS
XM_MN_EB NET202 EB NET206 NMOS
XM_MN_E NET190 E NET206 NMOS
XM_MN_D GND D NET190 NMOS

[Failed to route net CPB1]:
Failed to route net region: from x=42 to x=44
devices in the unrouted region:
XXITG_CP2/XI0/M_IPA D3 CPB1 D2 PMOS
XXITG_CPB1/XI0/M_IPA D5 CP2 D3 PMOS
XXITG_CP2/XI0/M_INA D3 CP2 D2 NMOS
XXITG_CPB1/XI0/M_INA D5 CPB1 D3 NMOS
Failed to route net region: from x=44 to x=56
devices in the unrouted region:
XXI_DB1/XI0/M_IPA VDD DB1 D2 PMOS
XXITG_CP2/XI0/M_IPA D3 CPB1 D2 PMOS
XM_MP1_PFB NET029 CPB1 DB1 PMOS
XM_MP0_PFB1 VDD D2 NET029 PMOS
XM_MP0_PFB0 VDD CDN NET029 PMOS
XM_MP_CP2 NET182 CP2 DB1 PMOS
XXI_DB1/XI0/M_INA GND DB1 D2 NMOS
XXITG_CP2/XI0/M_INA D3 CP2 D2 NMOS
XM_MN2_PFB NET0222 CP2 DB1 NMOS
XM_MN0_PFB GND CDN NET0181 NMOS
XM_MN1_PFB NET0181 D2 NET0222 NMOS
XM_MN_CPB1 NET182 CPB1 DB1 NMOS
```

Fig. 8: An example of extracted routability report prompt of Seq13 in Table I.

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10.5 A 12b 600MS/s Pipelined SAR and 2×-Interleaved Incremental Delta-Sigma ADC with Source-Follower-Based Residue-Transfer Scheme in 7nm FinFET

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Recently, pipelined ADCs are frequently adopted as sub-ADCs in multi-channel time-interleaved ADCs because a pipelined manner can reduce the number of interleaved channels by maximizing single-channel ADC operating speed. To reduce power consumption of residue amplifier, previous works [2,4,5] have explored power efficient residue amplifier architecture, such as dynamic amplifiers (DAs), open-loop amplifiers (OAs) and ring amplifiers (RAs). However, inter-stage gain error of power-efficient DAs, OAs, and RAs needs power-consuming gain calibration to compensate process, power supply, and temperature (PVT) variations.

This work presents a gain-calibration-free pipelined ADC architecture with the use of an open-loop residue-transferring buffer. The ADC consists of the 1st-stage successive-approximation (SAR) ADC, the 2nd-stage delta-sigma modulator (DSM), and a residue buffer. Instead of amplifying residue signal, the residue buffer transfers residue signal to the 2nd stage for pipelined operation. The non-amplified small residue signal is resolved by the 2nd stage DSM, which has wide dynamic range. A capacitor DAC (CDAC) in the 1st stage and a residue buffer are replicated in the feedback path of the 2nd stage DSM to eliminate PVT-variation-oriented inter-stage gain error without gain calibration. In addition, duration-fixed synchronous-like asynchronous clock, adaptive speed-controlled (ASC) clock, guarantees robust operation of the SAR against PVT variations. At 300MS/s, the gain-calibration-free ADC achieves a peak SNDR of 58.2dB. Moreover, the ADC has a 55.3dB SNDR with a 1Vpp swing, 500MHz (beyond Nyquist) input at 600MS/s. The ADC has superior speed over a wide input bandwidth compared with a sub-ranging SAR/DSM [1] and a noise-shaping pipelined SAR [3].

The ADC architecture and the corresponding timing diagram are shown in Fig. 10.5.1. The proposed ADC consists of a 7b single-channel SAR 1st stage, a 6.5b 2-channel time-interleaved 2nd-order continuous-time (CT) incremental DSM 2nd stage, and a residue buffer. The 1.5b redundancy can relax an inter-stage offset error requirement. The 1st stage samples the input signal with bootstrapped bottom-plate sampling. The sampling capacitance is 690fF to meet kT/C noise and residue signal accuracy. In the conversion phase, the 1st stage SAR resolves 7 bits with equally periodic 7 cycles of internal operation clock, CK_{ASC}, which triggers the comparator and SAR logic. After 7b conversion, the residue signal in a CDAC of the 1st stage is transferred to the 2nd stage by an open-loop residue buffer. The gain error of the residue signal path causes inter-stage gain error, which comes from the parasitic capacitance of a CDAC in the 1st stage and a gain error of open-loop residue buffer. The prototype ADC eliminates inter-stage gain error by matching the 2nd stage feedback path gain to residue signal path gain, rather than adopting digital calibration. The 2nd-stage CT incremental DSM operates at 300MS/s with 14 cycles of a 4.8GHz operation clock, CK_{DSMC}, which is buffered and enveloped from an externally applied CK_M. The 2-channel DSM operates in a time-interleaved manner to achieve a 600MS/s sampling rate. The linearity requirement of the time-interleaved 2nd stage is relaxed because the 2nd stage samples timing skew-less residue signal. As a result, the 2nd stage requires only offset calibration between interleaved ADCs.

The operating principle of the proposed architecture is described by the equation in Fig. 10.5.2. Quantization noise (Q-noise) consists of a gain mismatch term and a noise-shaping term. The noise-shaping term can be minimized under sufficiently high loop gain, H(s). Also, the gain mismatch term can be minimized by eliminating gain mismatch between the residue signal path and the feedback path of the 2nd stage. Fig. 10.5.2 shows the circuit implementation of residue buffer and the 2nd stage single-channel DSM. The 2nd-order noise-shaping DSM with 14 cycles of conversion clock is a proper choice in the 2nd stage to increase H(s), and it results in 6.5b resolution including 1.5b redundancy. Although the Gm-C integrator consumes less power, an active-RC integrator is widely used in CT DSM because the Gm-C integrator has a narrow linear input range. However, the coarse-converted residue signal enables the DSM to apply a power-efficient Gm-C integrator. A simple open-loop source follower is applied as a residue buffer. Since the source follower is robust to PVT variations and easy to implement, the design complexity can be relaxed. A CDAC and a residue buffer are replicated in the 2nd stage in Fig. 10.5.2 to eliminate the inter-stage gain error. For better gain matching, a comparator of the 1st stage is also replicated in the feedback path as a dummy. Well-matched replica circuits with shared-bias and adjacent layout arrangements remove inter-stage gain error from chip-to chip gain variation of the residue signal path. Moreover, the area-consuming 7b CDAC can be used in calibrating inter-stage offset error by adding offset in the feedback signal. Therefore, the comparator without offset calibration enables the SAR to enhance conversion speed.

Figure 10.5.3 shows the proposed adaptive speed-controlled (ASC) clock generation scheme. The ASC clock generator takes advantages from both the synchronous and asynchronous schemes by generating a 50% duty and duration-fixed internal operation clock (CK_{ASC}) cycles, regardless of the PVT variations, by background frequency calibration. The ASC clock generator dedicates at least a 1/2 clock period of CK_{ASC} for the internal DAC settling time. In fact, the DAC settling initiates as soon as the comparator makes the decision (just like the asynchronous operation) that dedicates more than ½ the clock period of CK_{ASC} for additional performance enhancement. At the end of the data conversion, a shift-register in the SAR logic counts the number of CK_{ASC} cycles and generates the end-of-conversion (EOC) signals. As shown in Fig. 10.5.3, the ASC clock generator calibrates the frequency of CK_{ASC} in the background by reducing or increasing the delay of the clock generator. The ASC clock generator consists of an inverter and capacitor-based ring oscillator and the delay line controller. Since the input sampling is performed by low-jitter CK_{SAMP}, the delay-controlled ring oscillator only cares to generate seven CK_{ASC} cycles with a relaxed jitter requirement for SAR operation. In addition, the duration-fixed synchronous-like asynchronous ASC clock helps the ADC easily detect the comparator meta-stability like a synchronous SAR.

The prototype ADC, implemented in 7nm FinFET CMOS technology, occupies 0.037mm^2 as shown in Fig. 10.5.7. The foreground on-chip offset calibration is performed to match the signal range between the 1st and the 2nd stage. Moreover, off-chip offset calibration is done between the 2-channel DSMs. No inter-stage gain calibration is applied with the help of the feedback path replica circuits. At 600MS/s under a 0.8V supply, the total power consumption is 13mW. The residue source follower consumes only 0.7mW. Measured performance is shown in Fig. 10.5.4. The ADC operates over a wide range (300 ~ 600MS/s) of sampling rates, although it adopts a CT DSM. The ADC achieves a peak SNDR of 58.2dB and a peak SFDR of 79.4dB at 300MS/s with a 5MHz input. At 600MS/s, the peak SNDR is 56.6dB, and it degrades less than 1.3dB up to 500MHz input. The measured SNDR at 600MS/s stays above 54dB across a 0.77~0.9V power supply range and -40~80°C temperature variations. Figure 10.5.5 illustrates the measured spectra and DNL/INL. The achieved DNL and INL are 0.71 and 1.45LSB, respectively. Figure 10.5.6 compares the ADC performance with the state-of-the-art ADCs. The proposed gain calibration-free ADC shows outstanding performance beyond the Nyquist frequency with only 1Vpp input swing.

References:

[1] A. Venca et al., "27.8 A 0.076mm2 12b 26.5mW 600MS/s 4×-interleaved subranging SAR-ΔΣ ADC with on-chip buffer in 28nm CMOS," *ISSCC*, pp. 470-472, Feb. 2016.
[2] B. Hershberg et al., "3.6 A 6-to-600MS/s Fully Dynamic Ringamp Pipelined ADC with Asynchronous Event-Driven Clocking in 16nm," *ISSCC*, pp. 68-70, Feb. 2019.
[3] Y. Song et al., "9.6 A 2.56mW 40MHz-Bandwidth 75dB-SNDR Partial-Interleaving SAR-Assisted NS Pipeline ADC With Background Inter-Stage Offset Calibration," *ISSCC*, pp. 164-166, Feb. 2020.
[4] J. Lagos et al., "A single-channel, 600Msps, 12bit, ringamp-based pipelined ADC in 28nm CMOS," *IEEE Symp. VLSI Circuits*, pp. C96-C97, June 2017.
[5] K. Moon et al., "A 9.1 ENOB 21.7fJ/conversion-step 10b 500MS/s single-channel pipelined SAR ADC with a current-mode fine ADC in 28nm CMOS," *IEEE Symp. VLSI Circuits*, pp. C94-C95, June 2017.

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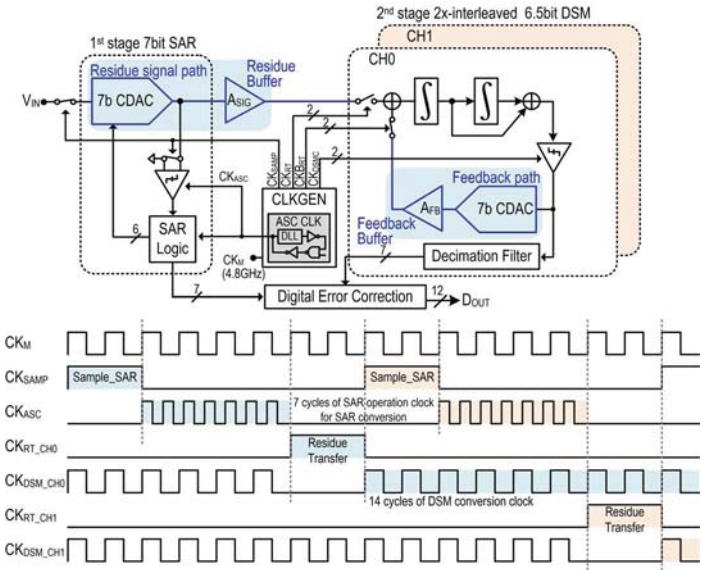


Figure 10.5.1: Block and timing diagram of the prototype 12b pipelined SAR and 2×-interleaved CT incremental DSM.

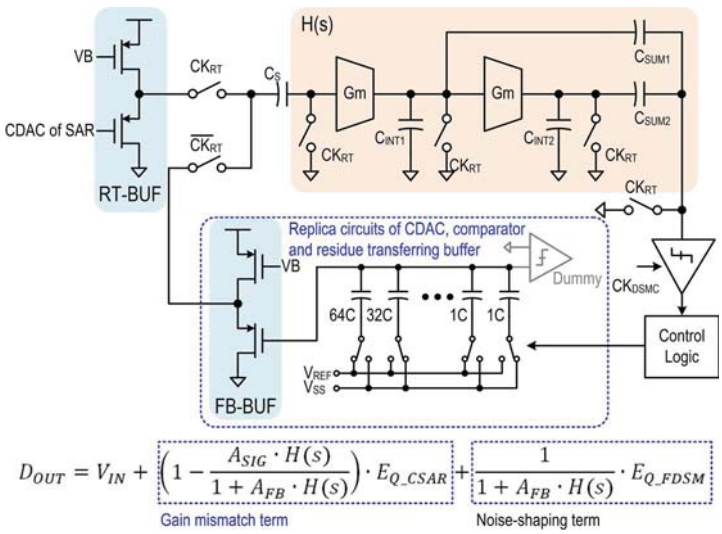


Figure 10.5.2: The circuit implementation of the 2nd stage and the operation principle of gain-error-calibration-free residue transfer scheme.

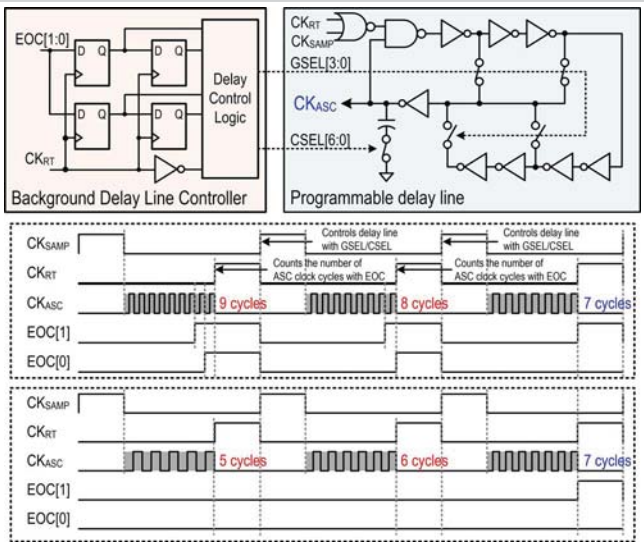


Figure 10.5.3: The circuit implementation of the adaptive speed-controlled (ASC) clock generator in the 1st stage (Top) and timing diagram of background calibration by increasing (middle) or reducing (bottom) delay.

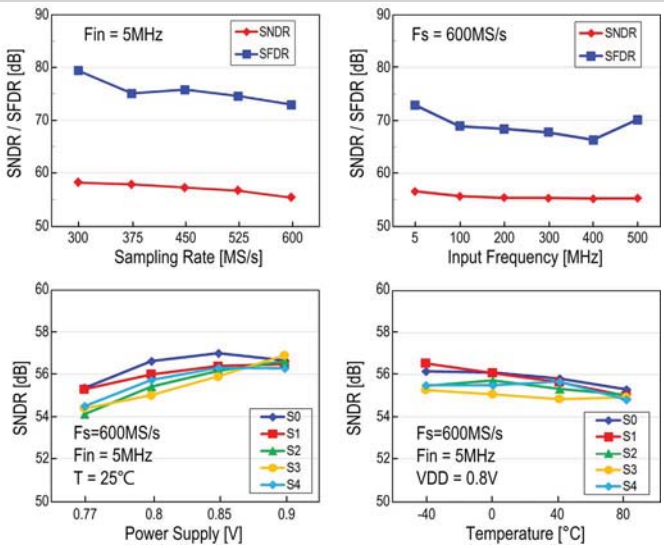


Figure 10.5.4: Dynamic performance and measured SNDR across power supply (0.77V ~ 0.9V) and temperature (-40 to 80°C) variations of 5 randomly selected chips at 600MS/s.

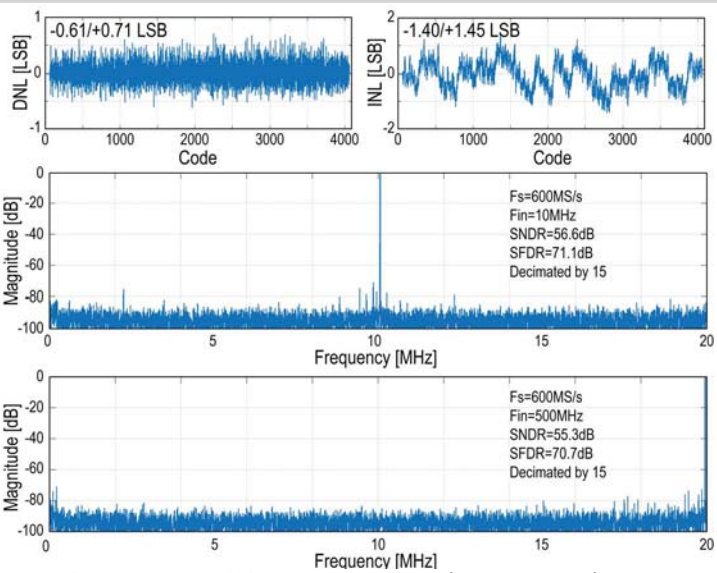


Figure 10.5.5: Measured 16384-point output spectra (decimated by 15) and DNL/INL at 600MS/s.

	This work	[1] ISSCC2016	[2] ISSCC2019	[4] VLSI2017	[5] VLSI2017
Architecture	**Pipe-SAR &TI DSM**	TI subranging SAR&DSM	Pipeline	Pipeline	Pipe-SAR
Technology	**7nm**	28nm	16nm	28nm	28nm
Resolution [bits]	**12**	12	11	12	10
Sample Rate [MS/s]	**600**	600	600	600	500
Supply Voltage [V]	**0.8**	1.2/1.5	0.85	0.9	1
Input swing [$V_{pp,diff}$]	**1**	1.5	1.5	1.6	1.6
Input Bandwidth [MHz]	**500**	265	600		250
SFDR @ Nyq. [dB]	**67.8**	-	78.3	69.2	69.2
SNDR @ Nyq. [dB]	**55.3**	58	60.2	56.3	56.6
Power [mW]	**13**	26.5	6	14.2	6
FoM_{Walden} @ Nyq. [fJ/c-s]	**45.6**	68	12	44.3	21.7
$FoM_{Schreier}$ @ Nyq. [dB]	**158.9**	158	167.2	159.5	162.8
Area [mm²]	**0.037**	0.076	0.037	0.62	0.015
Gain Calibration	**X**	O	O	O	O

Figure 10.5.6: Comparison with state-of-the-art.

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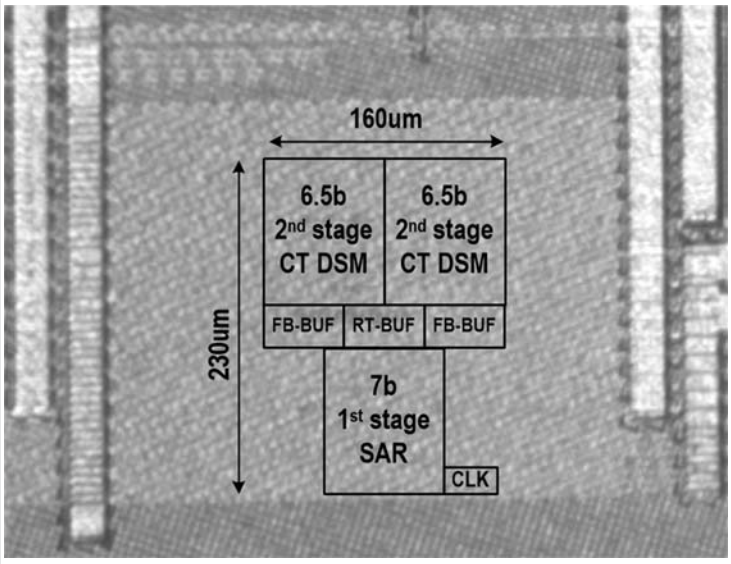


Figure 10.5.7: Die photo.

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